

Optimization for Data Science

Module 3: Constrained Optimization

Lecture 5: Introduction to Constrained Optimization

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Outline

- 1 Motivation and Recap
- 2 Basic Concepts of Constrained Problems
- 3 Data Science Examples
- 4 Preview of Methods
- 5 Conclusion

Why Constrained Optimization Matters

Practical Context in Data Science:

- Many real problems require that solutions obey certain bounds or relationships.
- We cannot freely choose parameters; they must lie in a **feasible set**.
- Examples:
 - Resource allocation with limited budgets.
 - L1 or L2 norm bounds for regularization.
 - SVM margin constraints in classification.



max. $f(x)$
↑

max.
Resource constraints!

Bank-balance

∞

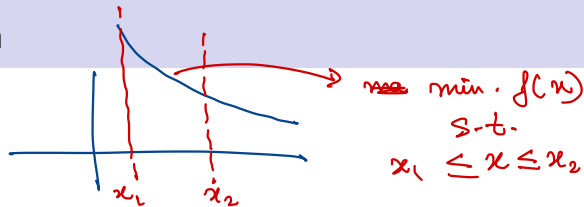
S.t.:

Difference from Unconstrained:

- Previously, we minimized $\min f(x)$ with no restrictions on x .
- Now, we have $\min f(x)$ subject to constraints like $g(x) \geq 0$ or $h(x) = 0$.
- The best solution may lie on a boundary or corner.

$g(x) \geq 0$

Recap: Unconstrained Optimization



What We Learned Before:

- Methods such as gradient descent, Newton's method, and line search for choosing step sizes.
- These rely on setting $\nabla f(\mathbf{x}) = \mathbf{0}$ for stationary points.

Now with Constraints:

- We cannot just find where $\nabla f(\mathbf{x}) = 0$, because $\mathbf{x}$ must remain feasible.
- Constraints may force the optimum onto boundaries where $\nabla f(\mathbf{x}) \neq 0$ in the usual sense.

Lesson:

- We need additional theory (e.g., Lagrange multipliers, KKT).
- We also need specialized methods to ensure solutions stay in the feasible region.

$w_1, w_2, w_3 \rightarrow$

SVM | Lasso | Ridge
Regularization in
D.S. ...

General Formulation

The Canonical Problem:

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}) \quad \text{subject to} \quad g_i(\mathbf{x}) \leq 0 \quad (i = 1, \dots, m), \quad h_j(\mathbf{x}) = 0 \quad (j = 1, \dots, p).$$

Feasible Set:

- The set of $\mathbf{x}$ that satisfy all inequality and equality constraints.
- We only search for minima within this region.

Why Harder Than Unconstrained:

- The unconstrained optimum might lie outside the feasible set.
- We must account for boundaries or surfaces defined by constraints.

$$7/30$$

$$1 = 3y^2 \quad x^2 = 1 - \frac{1}{3}$$

$$y = \pm \sqrt{\frac{1}{3}} \quad x = \sqrt{\frac{2}{3}}$$

$$\left[\begin{array}{l} \text{max.} \rightarrow (1-y^2)(y) \\ y - y^3 \quad 1 - 3y^2 = 0 \end{array} \right]$$

$$\begin{array}{l} \text{max. } x^2 y \\ \text{s.t.} \\ x^2 + y^2 = 1 \end{array}$$

$$0.125$$

$$x^2 = 1 - y^2 \Leftrightarrow$$

$$\frac{2}{3\sqrt{3}}$$

$$x = \sqrt{\frac{2}{3}} \\ y = \sqrt{\frac{1}{3}}$$

$$\checkmark \text{max } x^2 y \\ \left(\frac{2}{3}\right) \left(\sqrt{\frac{1}{3}}\right)$$

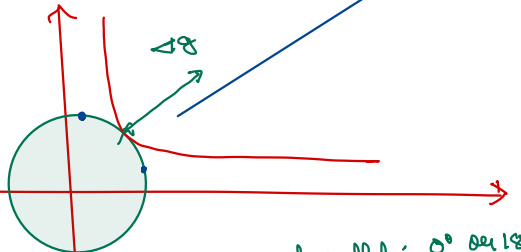
$$x \in \mathbb{R} \\ y \in \mathbb{R}$$

$$f = x^2 y$$

$$g = x^2 + y^2 = 1$$

$$\nabla f = \begin{bmatrix} 2xy \\ x^2 \end{bmatrix}$$

Example



$$\nabla g = \begin{bmatrix} 2x \\ 2y \end{bmatrix} \quad \nabla f - \lambda \nabla g = 0$$

$$\begin{bmatrix} 2xy \\ x^2 \end{bmatrix} - \lambda \begin{bmatrix} 2x \\ 2y \end{bmatrix} = 0$$

parallel: 0° or 180°

$\nabla g \propto \nabla f$
 $\nabla g = \lambda \nabla f$
 Lagrange multiplier!

$$\min. f(\vec{x})$$

st.

$$\boxed{g(\vec{x}) = 0}$$

$$\mathcal{L} = \underbrace{f(\vec{x}) + \lambda g(\vec{x})}_{\text{Informed by constraint}} \leftarrow \frac{\partial \mathcal{L}}{\partial \lambda} = g(\vec{x})$$



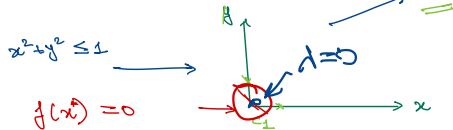
$$\frac{\partial \mathcal{L}}{\partial x} = 0; \quad \frac{\partial \mathcal{L}}{\partial y} = 0, \dots \quad \frac{\partial \mathcal{L}}{\partial \lambda} = 0$$

$$\boxed{g(\vec{x}) = 0}$$

min. $x^2 y$
s.t.

$x^2 + y^2 \leq 1$
 $x \geq 0$
 $y \geq 0$

3 constraints



$= \rightarrow \leq$

{ Active
Inactive }

Region
of feasibility

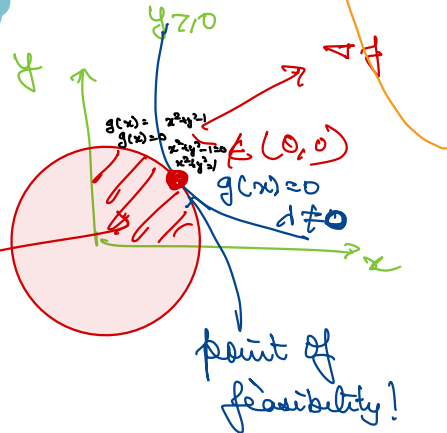
How if:
 $x = 55$

$x \geq 55$
How if:

(99)
fill

max. $x^2 y$
s.t.

$x^2 + y^2 \leq 1$
 $x \geq 0$
 $y \geq 0$



$$\begin{array}{ll}\min & x^2 y \\ \text{s.t.} & x^2 + y^2 \leq 1\end{array}$$

$$\mathcal{L} = x^2 y + \lambda (x^2 + y^2 - 1)$$

$$\frac{\partial \mathcal{L}}{\partial x} = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0$$

$$\begin{array}{ll}\min & -x^2 y \\ \text{s.t.} & x^2 + y^2 \leq 1\end{array}$$

$$\mathcal{L} = -x^2 y + \lambda (x^2 + y^2 - 1)$$

$$\frac{\partial \mathcal{L}}{\partial x} = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0$$

K-K-T-Conditions!

↓

Karush Kuhn Tucker!

If your constraints are inequality type,
In addition to $\nabla(\text{Lagrange}) = 0$
[primal]; some dual conditions are
also needed to be satisfied...

$$\underline{\lambda} \underline{g(x)} = \underline{0};$$

$$\begin{aligned} \min. & f(x) \\ \text{s.t.} & g(x) \leq 0 \\ \mathcal{L} &= f(x) + \lambda g(x) \\ & \lambda \geq 0 \end{aligned}$$

Active and Inactive Constraints

Definition:

- **Active** if constraint is exactly satisfied (e.g., $g_i(\mathbf{x}) = 0$) at the solution.
- **Inactive** if the solution strictly satisfies $g_i(\mathbf{x}) < 0$, so it does not touch the boundary.

Example: Box Constraints

- Suppose $x \in [a, b]$.
- If the optimum is inside (a, b) , neither boundary is active.
- If the optimum is below a , the constraint $x \geq a$ is active.

Consequences:

- Identifying which constraints are active can be important in forming the final conditions for optimality.

LR Clustering

Regularized Learning!

SUR / SUM

$$\rightarrow \min. f(x)$$

$$\rightarrow \equiv \max. -f(x)$$

- ① GD ++
- ② Newton ++
- ③ Heuristics → something
- ④ $\nabla f = 0$
 $\nabla^2 f$: check

$$\min. f(x, y)$$

s.t.

$$g(x) \leq 0$$

$$x \geq 0$$

$$\Rightarrow -x \leq 0$$

$$\min. f(x, y) + \lambda_1 g(x, y) + \lambda_2 (-x)$$

$$\lambda_1 = 0$$

$$\lambda_2 = 0$$

NN
DL++

$$\left\{ \begin{array}{l} \min. f(x) \\ \text{s.t.} \\ g(x) = 0 \end{array} \right\}$$

$$\mathcal{L} = f(x) + g(x)x$$

"point" solution
 or "penalty" method

$$\min. f(x)$$

s.t.

$$g(x) \leq 0$$

$x_1 \rightarrow$ unbounded
 $x_2 \rightarrow$ bounded

$$\min. f(x)$$

s.t.

$$g(x) \leq 0$$

Lagrange
 +
 KKT

$$\min. f(x)$$

s.t.

$$g(x) \geq 0$$

$$\min. f(x)$$

s.t.

$$-g(x) \leq 0$$

$$\min. f(x)$$

s.t.

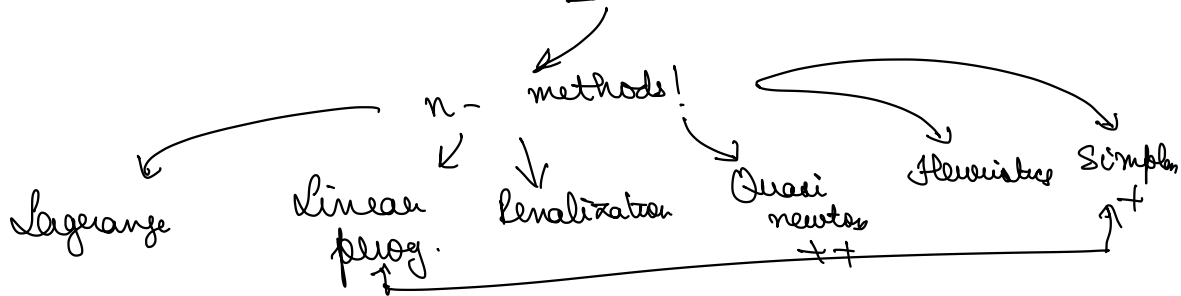
$$-g(x) \geq 0$$

$$h(x) \leq 0$$

$$\underline{d_1}, \underline{d_2}$$

necessary!

But
not efficient!!

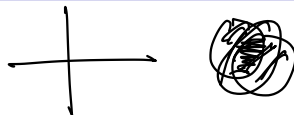


Integer Optimization!

Stationarity Revisited

Unconstrained Case:

$$\nabla f(\mathbf{x}^*) = \mathbf{0}.$$



Constrained Case (Preview):

$$\nabla f(\mathbf{x}^*) + (\text{combination of constraint gradients}) = \mathbf{0}.$$

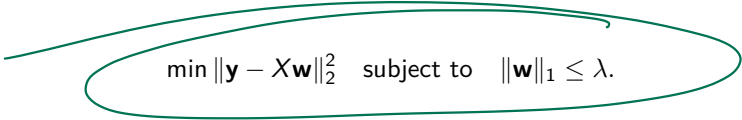
- If a constraint is inactive, it does not affect the solution.
- If it is active, its gradient might appear in the stationarity condition (via multipliers).

Next Lectures:

- We will introduce Lagrange multipliers and KKT conditions that unify these ideas.

Example 1: L1-Bounded (Lasso)

Form:


$$\min \|\mathbf{y} - X\mathbf{w}\|_2^2 \quad \text{subject to} \quad \|\mathbf{w}\|_1 \leq \lambda.$$

Interpretation:

- We want a solution $\mathbf{w}$ that fits the data but also remains within an L1 ball of radius λ .
- Geometrically, the feasible set is a diamond shape in coefficient space.

Data Science Reasoning:

- Enforcing sparsity: many coefficients become zero if λ is small.
- Alternatively viewed as unconstrained with a penalty $\alpha \|\mathbf{w}\|_1$, but constraints show the direct geometric meaning.

Example 2: Non-negativity Constraints

Scenario:

- We might want $\mathbf{x} \geq \mathbf{0}$ in certain tasks (resource usage, certain factorization models).

Feasible Set:

$$\{\mathbf{x} \in \mathbb{R}^n \mid x_i \geq 0 \quad \forall i\}.$$

- This is the positive orthant in $\mathbb{R}^n$.

Consequence:

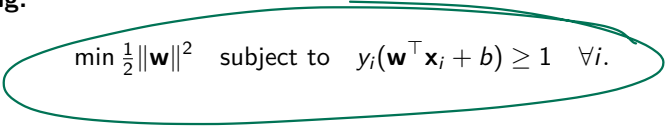
- If unconstrained optimum tries to pick negative values, we clamp at 0 in the final solution.

Example in DS:

- Non-negative matrix factorization for images or signals.
- We must remain feasible in a domain that does not allow negative parameter values.

Example 3: SVM Margin Constraints (Sneak Peek)

Classification Setting:


$$\min \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{subject to} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 \quad \forall i.$$

Why Constraints:

- Each data point must lie on the correct side of margin boundary.
- We maximize margin by minimizing $\|\mathbf{w}\|$, but only if constraints hold.

Relation to Constrained Optimization:

- This is a classical linear inequality setup.
- Slack variables allow “soft margin” if perfect separation is not possible.

Methods to Handle Constraints (Preview)

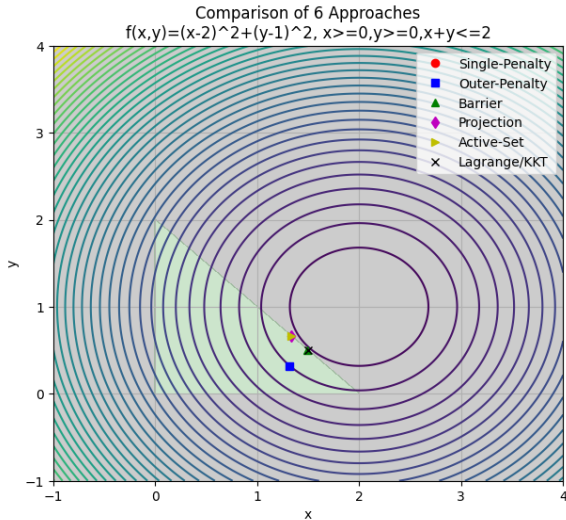
Approaches:

- **Penalty Methods:** Add large cost for violating constraints.
- **Barrier or Interior-Point:** Keep solutions inside feasible region by infinite cost near boundary.
- **Projection or Proximal:** For simple sets (e.g., box constraints), project each update onto feasibility.

Deeper Details in Lecture 6:

- We will introduce each method's intuition, basic math, and interpretation.
- Show a small numeric example to illustrate how solutions remain feasible.
- Data science perspective: which approach is common in large-scale or structured problems.

Preview of Outer Penalty Results



Lagrange Multipliers and KKT Conditions (Brief Mention)

Why We Need Them:

- In unconstrained: $\nabla f = 0$.
- In constrained: we balance ∇f with constraint gradients, scaled by multipliers.

KKT Conditions:

- Generalize stationarity, feasibility, dual feasibility, and complementary slackness.
- Provide a systematic framework for analyzing constrained optima.

When We Will Cover:

- In a future lecture (Lecture 6 or 7), we dive into the full math.
- Then connect it to SVM or lasso for a deeper view of how constraints shape the final solution.

Takeaways from Lecture 5

Main Points:

- Constrained optimization introduces a **feasible region** that must be respected.
- Active vs. inactive constraints can push the optimum onto boundaries or corners.
- Several data science problems (lasso, SVM, bounding parameters) are fundamentally constrained.

Looking Ahead to Lecture 6:

- We will learn how to solve constrained problems:
 - Penalty/barrier approaches.
 - Projection methods for simple constraints.
 - Understanding the trade-offs and real implementations.
- We will also introduce more theory (Lagrange multipliers, KKT) soon.

Thank you!